

AI-Driven Automated Metasurface Design Framework for Diverse RCS Reduction Objectives

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Abstract—This paper presents an AI-driven automated design framework that directly maps prescribed monostatic radar cross section reduction (RCSR) objectives to the final designs within a unified design process. It avoids the repeated expert-dependent multi-step redesign generally required by conventional approaches when the prescribed objectives change. A convolutional neural network (CNN)-based surrogate model is constructed for fast electromagnetic response prediction of generated units, while a long short-term memory (LSTM)-based reinforcement learning policy sequentially generates the unit topologies and array arrangements according to the prescribed RCSR objective. Furthermore, mirror-symmetric unit-cell configurations and the corresponding paired arrangement are incorporated to provide polarization-insensitive characteristics of generated designs. Finally, two RCS reduction plates with different prescribed objectives (denoted as RCSR-MS1 and RCSR-MS2) are generated and validated through full-wave simulations, fabrication, and measurements. Under normally incident LP waves, RCSR-MS1 achieves 15-dB RCSR over 9.6–21.2 GHz, while RCSR-MS2 achieves 10-dB RCSR over 7.0–22.8 GHz (106.1% fractional bandwidth). Under normally incident CP waves, RCSR-MS1 achieves approximately 15-dB RCSR over 9.5–21 GHz, while RCSR-MS2 achieves approximately 10-dB RCSR over 7–23 GHz (106.7% fractional bandwidth). Moreover, the generated designs also exhibit a certain degree of angular stability, with the prescribed RCS reduction performance generally maintained within the incidence angles of approximately 30° . Both generated designs satisfy their respective prescribed RCSR objectives, thereby validating the effectiveness of the proposed framework and its requirement-to-complete-design capability for different RCS reduction objectives.

Index Terms—Radar Cross Section Reduction, Metasurface, AI-driven framework, Reinforcement Learning, Surrogate Model.

I. INTRODUCTION

RADAR cross section (RCS) reduction metasurfaces play a critical role in low-observable system design, electro-

magnetic compatibility, and stealth-related applications [1]–[4]. With the increasing complexity of electromagnetic environments, practical RCS reduction requirements have become increasingly diverse, involving different operating frequency bands and reduction levels. Accordingly, RCS reduction design has received increasing research attention in recent years.

Metasurfaces, as two-dimensional counterparts of metamaterials, have been widely employed for RCS reduction owing to their low profile, ease of integration, and flexible capability for electromagnetic-wave manipulation. Various designs of metasurface units have been developed to realize RCS reduction, including artificial magnetic conductors (AMCs) [5]–[14] and polarization conversion metasurfaces (PCMs) [15]–[20], among others. Despite their good RCS reduction performance, conventional RCS reduction metasurface designs generally require substantial expert guidance throughout a sequential and cumbersome multi-step design procedure, including unit-cell topology selection, repeated structural-parameter tuning or optimization, and subsequent array-arrangement design.

First, the unit-cell topology is generally selected or conceived according to the prescribed RCS reduction objective based on electromagnetic knowledge and previous design experience. In [7], metasurface elements consisting of a cross-dipole and three square rings are designed for 6-dB RCS reduction over 9.31–28.22 GHz. In [10], circular patches, square patches, and ring structures are combined to construct AMC elements with distinct reflection phases for achieving 10-dB RCS reduction over 3.75–10 GHz. In [17], a PCM element based on two 45° -oriented meander lines is proposed for 6-dB RCS reduction over 6.1–18.9 GHz by using the Pancharatnam–Berry (P–B) phase principle [21]. In [16], a PCM unit with complex metallic structures is designed for 15-dB RCS reduction over 15.4–34.3 GHz. These unit topologies are developed for different RCS reduction requirements, such as 6-dB, 10-dB, or 15-dB reduction over different operating frequency ranges. Different RCS reduction objectives may require different electromagnetic responses and different unit-cell topologies. Consequently, when the prescribed objective changes substantially, the originally selected topology may no longer provide the required response, and a new topology may need to be reconsidered or redesigned.

Second, after the unit-cell topology is determined, its geometrical parameters usually need to be repeatedly adjusted or optimized through full-wave simulations until the desired electromagnetic responses are obtained. To improve the efficiency of this process, artificial intelligence and machine-learning techniques have been introduced in [22]–[24] to replace or as-

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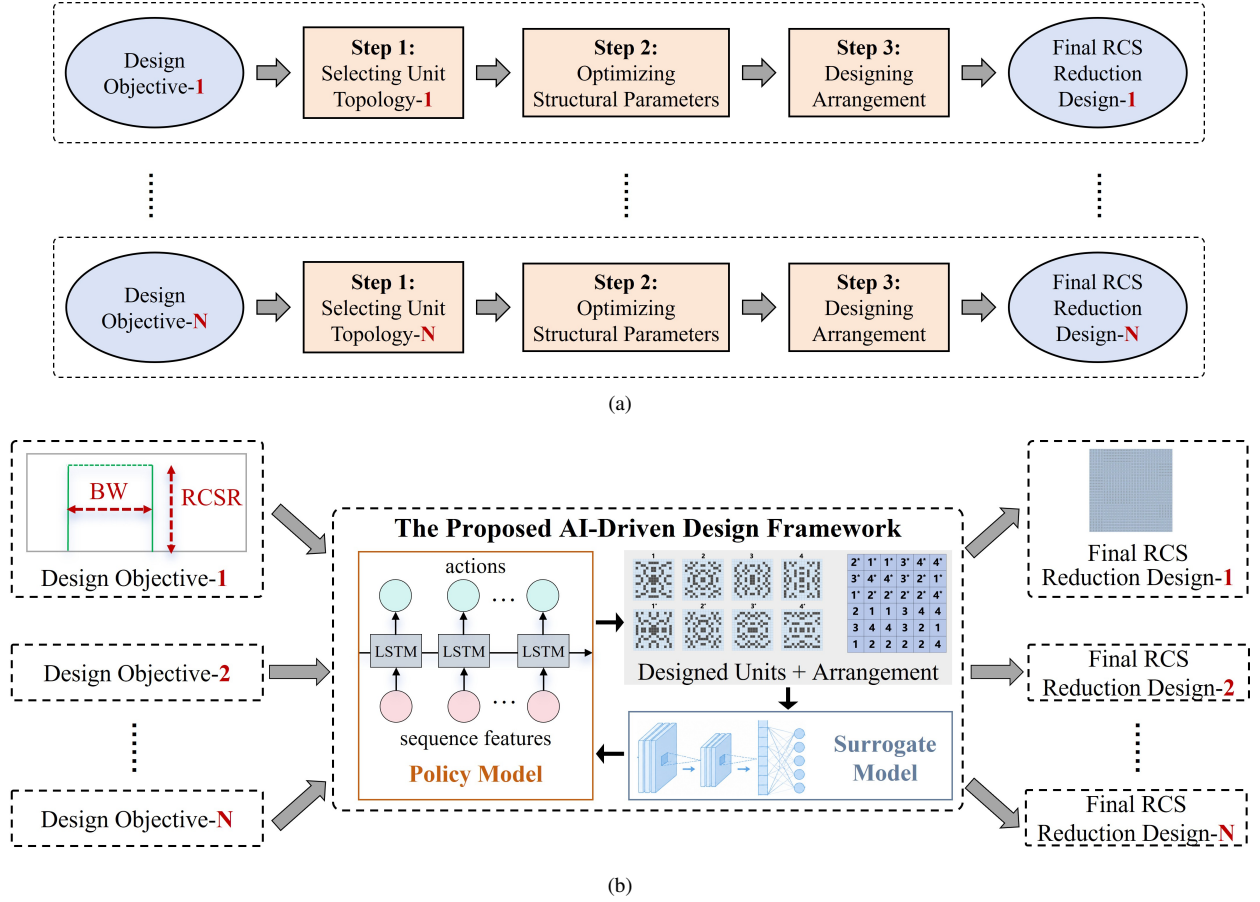


Fig. 1. Comparison of conventional design method and proposed framework. (a) Conventional multi-step design method. (b) Proposed requirement-driven AI-assisted design framework.

sist the conventional iterative parameter-optimization process for predefined unit-cell structures. However, these approaches generally operate within the parameter space of manually predefined unit-cell topologies and mainly optimize geometrical dimensions. Consequently, although the parameter optimization process can be significantly accelerated, the achievable design space remains constrained by the initially selected topology, and the overall design of electromagnetic responses still relies considerably on prior electromagnetic knowledge and expert-guided topology selection.

Finally, after suitable unit-cell responses are obtained, the arrangement of units over the plate must be further designed or optimized to achieve the prescribed RCS reduction performance. This step determines the specific spatial distribution, relative occupation, and combination of different unit cells over the final RCS reduction plate [14], [25]–[28]. In [25], the relative occupation of units over the plate is designed for improving monostatic RCS reduction performance. In [26], a concave/convex parabolic phase distribution is adopted to determine the unit arrangement. In [28], a cubic and vortex phase distribution is adopted to determine the unit arrangement. These studies demonstrate that the final RCS reduction performance depends not only on the electromagnetic responses of individual unit cells but also on their arrangement, making the array-level arrangement another essential design step following

unit topology selection and parameter optimization.

Therefore, the conventional design procedure can generally be summarized as "Expert-Guided Unit Topology Design → Structural Parameter Optimization → Arrangement Optimization". Different RCS reduction requirements may require different initial unit structures, parameter settings, and array configurations. Accordingly, when the prescribed RCS reduction objective changes, a substantial part of the above design procedure generally needs to be repeated as shown in Fig. 1(a). Moreover, conventional approaches remain highly dependent on expert experience and target-specific redesign. As a result, the above multi-step procedure often needs to be reconsidered for different RCS reduction objectives, and it remains insufficiently explored to develop a direct and unified approach for mapping different prescribed RCS reduction objectives to complete RCS reduction designs.

To address the above problems, this paper proposes a requirement-driven AI-assisted automated design framework for RCS reduction metasurfaces as shown in Fig. 1(b). The proposed framework establishes a direct mapping from prescribed RCS reduction objectives to complete metasurface designs, including both the required unit topologies and their corresponding array arrangements. A CNN-based surrogate model provides fast electromagnetic evaluation of the generated units, while an LSTM-based reinforcement learning

policy searches and generates the corresponding unit topologies and spatial arrangements according to the prescribed RCS reduction objective. In addition, mirror-symmetric unit configurations and the corresponding paired arrangement are incorporated into the proposed framework, thus improving the design efficiency and ensuring the polarization-insensitive characteristic of generated designs. To validate the proposed framework, two different RCS reduction objectives are prescribed, from which two distinct designs, RCSR-MS1 and RCSR-MS2, are automatically generated. Both designs are subsequently verified through simulations, fabrication, and measurements, and the obtained results confirm that they satisfy their respective prescribed RCS reduction objectives. The main contributions of this paper are summarized as follows:

1. Requirement-driven automated design framework:

A unified design framework is established to directly map prescribed RCS reduction objectives to complete designs. Different from conventional approaches based on predefined unit-cell topologies, the proposed framework directly searches and generates the unit topologies and their corresponding arrangements according to the prescribed objective. Therefore, it substantially reduces the dependence on expert-guided topology selection and repeated multi-step design, providing a more direct requirement-to-complete-design route for RCS reduction designs.

2. Adaptability to different prescribed RCSR objectives:

The proposed framework provides a unified design route for diverse RCS reduction objectives. For different prescribed objectives, it can directly search for and generate the corresponding complete designs, including new unit topologies and arrangements. This capability enables different objectives to be addressed within the proposed framework, avoiding repeated expert-guided and target-specific multi-step redesign in conventional methods. Two designs are generated for different prescribed RCSR objectives and both satisfy their respective design targets, demonstrating the adaptability of the proposed framework to different design objectives.

3. Favorable electromagnetic performance of the automatically generated designs:

The generated designs not only satisfy their respective prescribed RCS reduction objectives but also exhibit favorable electromagnetic characteristics. Owing to the mirror-symmetric structure of units and the corresponding paired arrangement, the generated designs exhibit polarization-insensitive RCS reduction performance and can maintain the prescribed objectives under both linearly polarized (LP) and circularly polarized (CP) normal incidences. Besides, they also exhibit a certain degree of angular stability, with the prescribed RCSR performance generally maintained for incidence angles up to approximately 30° . In particular, RCSR-MS2 achieves broadband 10-dB RCS reduction from 7.0 to 22.8 GHz (106.1% fractional bandwidth) using a single-substrate, 3-mm-low-profile configuration. These results demonstrate the effectiveness of the proposed framework while maintaining favorable electromagnetic performance of generated designs.

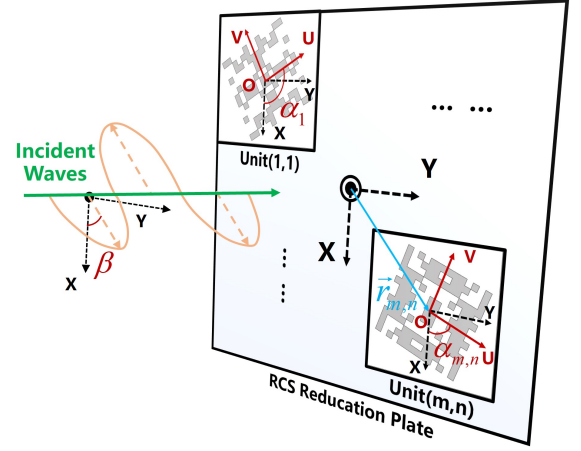


Fig. 2. Schematic diagram of monostatic RCS reduction plate.

II. THE PRINCIPLES AND METHODS OF RCS REDUCTION

Radar cross section (RCS) reduction can be quantified by comparing the scattered fields of the designed metasurface with those of a reference perfect electric conductor (PEC) with identical size:

$$\sigma_R = 10 \log_{10} \left(\frac{|E_{sca}^{PEC}|^2}{|E_{sca}^{Meta.}|^2} \right) \quad (1)$$

where $E_{sca}^{Meta.}$ and E_{sca}^{PEC} denote the scattered electric fields from the metasurface and the PEC plate, respectively.

Based on Eq.(1), a theoretical formulation is established to characterize the RCS reduction performance of a metasurface plate (c.f. Section II-A), where the overall scattered field is determined by the coherent superposition of all unit contributions. This indicates that the plate-level scattering performance is jointly determined by unit electromagnetic responses and their spatial arrangement. Accordingly, the overall design can be formulated as generating unit structures and array configurations to satisfy RCS reduction targets. Guided by this, an artificial intelligence (AI)-driven design framework is developed to efficiently generate metasurface plate designs for desired RCS reduction performance (c.f. Section II-B).

A. The Principle of Monostatic RCS Reduction

As shown in Fig. 2, an RCS reduction plate is composed of $M \times N$ metasurface units with a period of P , where m and n denote the element indices along the x - and y -directions, respectively. A sub-coordinate system UOV is established according to the structural symmetry axes of each metasurface unit, while the global coordinate system is denoted by XOY . For the unit (m, n) , the local coordinate system UOV is rotated from the global coordinate system XOY by an angle $\alpha_{m,n}$. For simplicity, the explicit frequency dependence is omitted in the following derivations, while the same formulation applies to all frequency points within the considered band. Therefore, the reflection coefficient matrix $\bar{R}^{UOV}$ in local coordinate system UOV and the position vector $\vec{r}$ in

global coordinate system XOY of the metasurface unit are given by

$$\left\{ \begin{array}{l} \overline{\overline{R}}_{m,n}^{UOV} = \begin{bmatrix} R_{m,n}^{uu} & R_{m,n}^{uv} \\ R_{m,n}^{vu} & R_{m,n}^{vv} \end{bmatrix}, \\ \vec{r}_{m,n} = (x_{m,n}, y_{m,n}, 0) \\ = (P(m - \frac{M+1}{2}), P(n - \frac{N+1}{2}), 0), \end{array} \right. \quad (2)$$

where $x_{m,n}$ and $y_{m,n}$ denote the coordinates of the unit center along the x - and y -directions. R^{uu} and R^{vv} denote the co-polarized reflection coefficients, and R^{uv} and R^{vu} denote the cross-polarized reflection coefficients. Since the local U/V axes are constructed according to the structural symmetry axes of the unit cell, the considered metasurface unit exhibits a non-polarization-conversion characteristic in this coordinate system, *i.e.*, $|R^{uv}| = 0$, $|R^{vu}| = 0$. Therefore, the reflection response of the unit exhibits the co-polarized characteristic.

The reflection response of a metasurface unit in the global coordinate system XOY is obtained by rotating its local reflection matrix from UOV to XOY , yielding

$$\left\{ \begin{array}{l} \overline{\overline{R}}_{m,n}^{XOY} = \begin{bmatrix} R_{m,n}^{xx} & R_{m,n}^{xy} \\ R_{m,n}^{yx} & R_{m,n}^{yy} \end{bmatrix}, \\ R_{m,n}^{xx} = R_{m,n}^{uu} \cos^2 \alpha_{m,n} + R_{m,n}^{vv} \sin^2 \alpha_{m,n}, \\ R_{m,n}^{xy} = (R_{m,n}^{vv} - R_{m,n}^{uu}) \sin \alpha_{m,n} \cos \alpha_{m,n}, \\ R_{m,n}^{yx} = (R_{m,n}^{vv} - R_{m,n}^{uu}) \sin \alpha_{m,n} \cos \alpha_{m,n}, \\ R_{m,n}^{yy} = R_{m,n}^{uu} \sin^2 \alpha_{m,n} + R_{m,n}^{vv} \cos^2 \alpha_{m,n}. \end{array} \right. \quad (3)$$

Under normal incidence of an electromagnetic wave E_{in} with arbitrary polarization, where the polarization direction makes an angle β with respect to the x -axis, the monostatic scattered field of the metasurface-based plate is derived as [25]:

$$\left\{ \begin{array}{l} E_{sca}^x = 2jk_0 \frac{e^{-jk_0 D}}{4\pi D} E_{in} \sum_{m=1}^M \sum_{n=1}^N (R_{m,n}^{xx} \cos \beta + R_{m,n}^{xy} \sin \beta) \\ E_{sca}^y = 2jk_0 \frac{e^{-jk_0 D}}{4\pi D} E_{in} \sum_{m=1}^M \sum_{n=1}^N (R_{m,n}^{yx} \cos \beta + R_{m,n}^{yy} \sin \beta), \end{array} \right. \quad (4)$$

where k_0 denotes the free-space wavenumber, and D is the distance between the observation point and the metasurface plate. For RCS evaluation under far-field observation, D is taken to approach infinity. The total monostatic scattered field amplitude of the metasurface-based plate is derived as

$$\left| E_{sca}^{Meta.} \right|^2 = \left| 2jk_0 \frac{e^{-jk_0 D}}{4\pi D} E_{in} \right|^2 \left(\left| \sum_{m=1}^M \sum_{n=1}^N (R_{m,n}^{xx} \cos \beta + R_{m,n}^{xy} \sin \beta) \right|^2 + \left| \sum_{m=1}^M \sum_{n=1}^N (R_{m,n}^{yx} \cos \beta + R_{m,n}^{yy} \sin \beta) \right|^2 \right) \quad (5)$$

Furthermore, since the co-polarized and cross-polarized reflection coefficients of PEC is -1 and 0, respectively, the total monostatic scattered field amplitude of PEC plate is

$$\left| E_{sca}^{PEC} \right|^2 = \left| 2jk_0 \frac{e^{-jk_0 D}}{4\pi D} E_{in} \right|^2 |MN|^2. \quad (6)$$

By substituting (3), (5), (6) into (1), the monostatic RCS reduction of metasurface-based plate can be derived as

$$\sigma_R = -10 \log_{10} \left(\frac{1}{|MN|^2} \left(\left| \sum_{m=1}^M \sum_{n=1}^N \left[(R_{m,n}^{uu} \cos^2 \alpha_{m,n} + R_{m,n}^{vv} \sin^2 \alpha_{m,n}) \cos \beta + (R_{m,n}^{vv} - R_{m,n}^{uu}) \sin \alpha_{m,n} \cos \alpha_{m,n} \sin \beta \right] \right|^2 + \left| \sum_{m=1}^M \sum_{n=1}^N \left[(R_{m,n}^{vv} - R_{m,n}^{uu}) \sin \alpha_{m,n} \cos \alpha_{m,n} \cos \beta + (R_{m,n}^{uu} \sin^2 \alpha_{m,n} + R_{m,n}^{vv} \cos^2 \alpha_{m,n}) \sin \beta \right] \right|^2 \right) \right) \quad (7)$$

The above formulation indicates that monostatic RCS reduction is jointly determined by the reflection responses of individual metasurface units and their spatial arrangement over the plate. Therefore, the overall scattering characteristics can be effectively controlled by jointly designing the unit reflection coefficients and the rotation-angle distribution. This also motivates the subsequent AI-driven framework, which jointly optimizes unit structures and array arrangement for the given RCS reduction targets.

B. AI-Driven Design Framework for RCS Reduction

Overview of the Framework. Based on the previous analytical formulation, the monostatic RCS reduction of a metasurface plate can be evaluated from the reflection responses of individual unit cells with their spatial arrangement. This provides a physical basis for automatic metasurface design: instead of directly performing repeated full-wave simulations for the entire metasurface plate, one can first predict the reflection responses of candidate unit cells and then substitute them into (7) to compute the overall RCS reduction performance.

Accordingly, the proposed framework aims to learn a direct requirement-to-design mapping, which can be written as

$$(\mathcal{U}, \mathbf{A}) = \mathcal{F}(\mathcal{B}, \sigma_{tar}), \quad (8)$$

where $\mathcal{B} = [f_{min}, f_{max}]$ denotes the target frequency band, σ_{tar} denotes the desired RCS reduction depth over $\mathcal{B}$, $\mathcal{U} = \{\mathbf{U}_1, \mathbf{U}_2, \dots, \mathbf{U}_K\}$ denotes the set of designed unit cells, and $\mathbf{A} \in \{1, \dots, K\}^{M \times N}$ denotes the arrangement matrix of the metasurface plate. Each entry of $\mathbf{A}$ specifies the index of the unit type assigned to the corresponding array position.

To facilitate automated generation, each unit is described by a pixelated binary topology matrix $\mathbf{U}_k \in \{0, 1\}^{Q \times Q}$, where $k = 1, \dots, K$. Here, $Q \times Q$ denotes the discretized design

Algorithm 1 Requirement-driven automated design of broadband RCS reduction metasurfaces.

Require: Target frequency band $\mathcal{B}$, target RCS reduction depth σ_{tar} , unit-cell dataset $\mathcal{D} = \{(\mathbf{U}_i, \mathbf{R}_i)\}_{i=1}^N$, surrogate model $g(\cdot; \mathbf{w})$, policy model $\pi(\cdot; \theta)$, sample budget L .

Ensure: Designed unit set $\mathcal{U}^*$ and array arrangement $\mathbf{A}^*$.

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1: // Data-driven surrogate model training
2: while not convergent do
3:   Sample a mini-batch  $\mathcal{D}_b \subset \mathcal{D}$ .
4:   Predict the responses  $\hat{\mathbf{R}}_i = g(\mathbf{U}_i; \mathbf{w})$ ,  $\forall \mathbf{U}_i \in \mathcal{D}_b$ .
5:   Update  $\mathbf{w}$  by minimizing the loss in Eq. (9).
6: end while
7: // Requirement-driven automated metasurface design
8: while not convergent do
9:   Sample the candidate  $\mathcal{U}, \mathbf{A} \sim \pi(\cdot; \theta)$ .
10:  Predict the responses  $\hat{\mathbf{R}}_i = g(\mathbf{U}_i; \mathbf{w})$ ,  $\forall \mathbf{U}_i \in \mathcal{U}$ .
11:  Compute RCSR curve  $\sigma(f)$  over  $\mathcal{B}$  via Eq. (7).
12:  Compute the reward  $r(\mathcal{U}, \mathbf{A})$  according to Eq. (10).
13:  Update  $\theta$  by optimizing the policy objective.
14: end while
15: Sample candidate designs  $\{(\mathcal{U}_j, \mathbf{A}_j)\}_{j=1}^L \sim \pi(\cdot; \theta)$ .
16: Obtain the final design  $(\mathcal{U}^*, \mathbf{A}^*) = \operatorname{argmax}_{(\mathcal{U}_j, \mathbf{A}_j)} r(\mathcal{U}_j, \mathbf{A}_j)$ .
17: Validate  $(\mathcal{U}^*, \mathbf{A}^*)$  by full-wave simulation and measurement.
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space, with an entry of 1 indicating the presence of metal and 0 indicating its absence at the corresponding location.

Within this framework, a surrogate model is first constructed to predict the unit-level reflection responses (*i.e.*, R^{uu} and R^{vv}). These predicted responses are then substituted into (7) to evaluate the overall RCS reduction. Based on this proxy evaluation, an LSTM-based policy model is employed to automatically generate the unit topologies and their array arrangement. The overall algorithm is presented in Algo. (1).

Surrogate Model for Unit Response Prediction. Let the frequency band be discretized as $\mathcal{B} = \{f_1, f_2, \dots, f_T\}$. Then, the response of the k -th unit can be written as $\mathbf{R}_k = [R_k^{uu}(f_1), \dots, R_k^{uu}(f_T), R_k^{vv}(f_1), \dots, R_k^{vv}(f_T)]$ where $R_k^{uu}(f_t)$ and $R_k^{vv}(f_t)$ denote the co-polarized reflection responses of the k -th unit at the frequency point f_t , respectively. A surrogate model is introduced to provide fast prediction of the unit-level electromagnetic responses. Specifically, each unit cell is represented by a binary topology matrix $\mathbf{U}_k \in \{0, 1\}^{Q \times Q}$, and the surrogate model is formulated as a deep neural network $\hat{\mathbf{R}}_k = g(\mathbf{U}_k; \mathbf{w})$, where $\hat{\mathbf{R}}_k$ denotes the predicted response, $g(\cdot; \mathbf{w})$ denotes the surrogate network, and $\mathbf{w}$ denotes its trainable parameters. In this work, $g(\cdot; \mathbf{w})$ is implemented as a convolutional neural network (CNN) [29].

Let $\mathcal{S} = \{(\mathbf{U}^{(i)}, \mathbf{R}^{(i)})\}_{i=1}^{N_s}$ denote the training dataset, where each sample consists of a unit topology matrix and its corresponding reflection responses obtained by full-wave simulation. The parameters $\mathbf{w}$ are learned by minimizing the discrepancy between the predicted and simulated responses over the dataset. The training objective is defined by

$$\mathcal{L}_{\text{sur}}(\mathbf{w}) = \frac{1}{N_s} \sum_{i=1}^{N_s} \left\| g(\mathbf{U}^{(i)}; \mathbf{w}) - \mathbf{R}^{(i)} \right\|_2^2, \quad (9)$$

where $\mathbf{R}^{(i)}$ includes both the U - and V -polarized reflection responses over all considered frequency points. The parameters $\mathbf{w}$ are optimized by gradient descent training.

LSTM-Based Policy Model for Design Generation. The metasurface design problem is formulated as a sequential generation process. To solve this process, a policy model π_θ is introduced, where π denotes the decision-making policy and θ its trainable parameters. π_θ is parameterized by an LSTM-based deep neural network [30], since the generation process is inherently sequential and each decision depends on the previously generated design states. Compared with Transformer- and diffusion-based networks, the LSTM-based policy network provides a more suitable balance between sequential dependency modeling, data efficiency, and computational cost for the discrete metasurface design problem.

In the design process, each unit cell $\mathbf{U}_k \in \{0, 1\}^{Q \times Q}$ is reshaped into a binary vector of length Q^2 , and the policy sequentially predicts the value at each location, with 1 indicating the presence of metal and 0 indicating its absence. This process is repeated for all K unit cells, yielding the complete unit set $\mathcal{U} = \{\mathbf{U}_1, \mathbf{U}_2, \dots, \mathbf{U}_K\}$. Subsequently, the arrangement matrix $\mathbf{A} \in \{1, \dots, K\}^{M \times N}$ is generated at each array position by assigning the index of the unit type, thereby yielding a metasurface design $\mathcal{D} = (\mathcal{U}, \mathbf{A})$. Solving the design problem means finding the parameters θ that define an effective policy, reinforcement learning [31] provides a framework for optimizing θ directly.

Once the metasurface design $\mathcal{D}$ is generated by the policy π , the surrogate model predicts the unit reflection responses, which are then substituted into (7) to evaluate the corresponding broadband RCS reduction performance. Let the target frequency band be discretized as $\mathcal{B} = \{f_1, f_2, \dots, f_T\}$, and σ_{tar} denote the desired RCS reduction performance. The reward is defined as the longest continuous frequency points where the achieved RCS reduction satisfies the target performance

$$\begin{cases} r = \max \left(\sum_{t=i}^j c_t \right), \\ c_t = \begin{cases} 1 & , \sigma_R(f_t) \geq \sigma_{\text{tar}}, \\ 0 & , \sigma_R(f_t) < \sigma_{\text{tar}}, \end{cases} \\ \forall t \in [i, j], \quad c_t = 1. \end{cases} \quad (10)$$

Here, $\sigma_R(f_t)$ denotes the evaluated RCS reduction at frequency point f_t . The binary variable c_t indicates whether the RCS reduction at f_t meets the prescribed target level, with $c_t = 1$ if $\sigma_R(f_t) \geq \sigma_{\text{tar}}$ and $c_t = 0$ otherwise. The reward r is then calculated as the maximum sum of consecutive c_t values, representing the longest contiguous frequency band over which the RCS reduction meets the prescribed target level. Note that in the design process, it may involve a long sequence of discrete decisions. However, the proposed reward is not a binary sparse signal that only indicates whether the final design satisfies the target requirement. Therefore, designs with different electromagnetic performances can receive different reward values. This reward formulation provides more informative optimization feedback and alleviates the sparse reward problem in long sequential generation.

To reduce reward variance and improve training stability, the raw reward is normalized in a group-wise manner, the

normalized reward is computed as

$$\tilde{r}_i = \frac{r_i - \mu_r}{\sigma_r + \epsilon}, \quad (11)$$

where r_i denotes the reward of the i th sampled design, μ_r and σ_r denote the mean and standard deviation of rewards within the group, respectively, and ϵ is a small constant for numerical stability. Based on the normalized reward, the policy model is optimized by a REINFORCE policy-gradient method. Through iterative policy updates, the policy is encouraged to produce unit topologies and arrangements that satisfy the target RCS reduction over as many frequency points as possible.

Once the policy model is trained, a set of candidate metasurface designs is sampled from the policy model, denoted as $\{(\mathcal{U}_j, \mathbf{A}_j)\}_{j=1}^L \sim \pi_\theta(\cdot)$, where L denotes the sampling budget. Each candidate is evaluated using the surrogate model-assisted reward function, and the design achieving the highest reward is selected as the final design:

$$(\mathcal{U}^*, \mathbf{A}^*) = \underset{(\mathcal{U}_j, \mathbf{A}_j)}{\operatorname{argmax}} r(\mathcal{U}_j, \mathbf{A}_j). \quad (12)$$

This selection ensures that the chosen metasurface maximizes the continuous frequency coverage of RCS reduction according to the target specifications. Finally, the selected design $(\mathcal{U}^*, \mathbf{A}^*)$ is validated through full-wave electromagnetic simulations and measurements, confirming the effectiveness of the reinforcement learning-based generation framework in producing high-performance metasurface designs.

III. DESIGNS FOR DIVERSE RCS REDUCTION TARGETS

Based on the proposed requirement-driven design framework, two designs, denoted as RCSR-MS1 and RCSR-MS2, are generated according to different RCSR objectives.

A. Configuration of RCS reduction plate

The complete RCS reduction plate consists of a 6×6 grid, corresponding to 36 sub-chessboard positions. Each sub-chessboard is physically implemented by a 5×5 block of identical metasurface unit cells with a period of 8 mm. Therefore, the final metasurface contains 30×30 units.

At the array-design level, only 18 of the 36 sub-chessboards need to be independently determined by the policy model. The remaining 18 sub-chessboards are automatically constructed using the corresponding 90° -rotated counterparts according to the predefined paired-arrangement strategy. Therefore, the 18-step arrangement generated by the LSTM-based policy is sufficient to determine the complete 6×6 sub-chessboard configuration. The 18 sub-chessboards are populated by 4 different metasurface unit topologies with different occupation ratios, forming a complete destructive phase surface in the monostatic direction for RCS reduction.

In the global coordinate system XOY , according to (3), a 90° rotation exchanges the two orthogonal co-polarized reflection responses of each generated unit ($R_{90^\circ}^{xx} = R_{0^\circ}^{yy}$, $R_{90^\circ}^{yy} = R_{0^\circ}^{xx}$). The paired arrangement of the generated units and their 90° -rotated counterparts therefore provides the electromagnetic basis for the polarization-insensitive characteristics of the complete RCS reduction plate. Meanwhile,

more phase differences are introduced, including those within each unit through rotating and those between different units, facilitating the generation of a destructive phase surface and simplifying the design process.

Each sub-chessboard is constructed by repeating the designed metasurface unit in a 5×5 arrangement of identical units. The 5×5 configuration in each sub-chessboard is adopted to provide a locally periodic electromagnetic environment that better approximates the periodic boundary conditions employed in the unit-cell simulations for surrogate-model training. Compared with directly placing isolated units, such local repetition reduces the discrepancy between the unit responses characterized under periodic boundary conditions and those realized in the finite plate.

B. Configuration of Metasurface Unit

As shown in Fig. 3 (a), the metasurface unit consists of a top metal structure and a substrate (thickness = 3 mm, dielectric constant = 2.2) backed by a perfect electric conductor (PEC). The top metal structure is symmetrically pixelated to enhance design flexibility and efficiency, overcoming the limitations of artificially designed metasurface structure. The metal structure is divided into a 16×16 pixel grid, with each pixel measuring $0.5 \text{ mm} \times 0.5 \text{ mm}$. In this binary matrix, "0" represents air and "1" represents PEC. To reduce coupling between metasurface units while maintaining performance stability, the outermost pixels are set to "0," creating a gap between adjacent unit cells. Thus, the effective design grid consists of 14×14 pixels.

Moreover, a local UOV coordinate system is defined along the two symmetry axes of the unit and the 14×14 grid is further divided into four regions, each containing a 7×7 pixel array. Only the 7×7 pixels in one quarter of the unit are independently generated by the policy model. The remaining three quarters are constructed through mirror symmetry with respect to the U- and V-axes. Therefore, although the generated metallic patterns may exhibit irregular pixelated geometries, all generated units satisfy predefined and physically meaningful symmetry constraints. This strategy simultaneously reduces the topology-search space while preserving sufficient structural diversity for electromagnetic-response design.

Owing to the mirror-symmetric configuration with respect to the local U- and V-axes, the generated metasurface units exhibit non-polarization-conversion characteristics under normally incident U- or V-polarized waves. Specifically, the magnitudes of cross-polarized reflection coefficients remain close to 0 ($|R^{uv/vu}| = 0$), while the magnitudes of the corresponding co-polarized reflection coefficients remain close to 1 ($|R^{uu/vv}| = 1$). Meanwhile, the co-polarized reflection phases (φ^{uu} and φ^{vv}) are generally different because of the anisotropic current distribution introduced by the generated topology. Therefore, the co-polarized reflection coefficients of the proposed unit cells are primarily determined by their corresponding reflection phases ($R^{uu} = e^{j\varphi^{uu}}$, $R^{vv} = e^{j\varphi^{vv}}$). Fig. 3 (b) presents the simulated surface-current distributions at 15 GHz for a representative generated mirror-symmetric unit under normally incident U- and V-polarized waves. Under the two orthogonal excitations, the induced surface currents

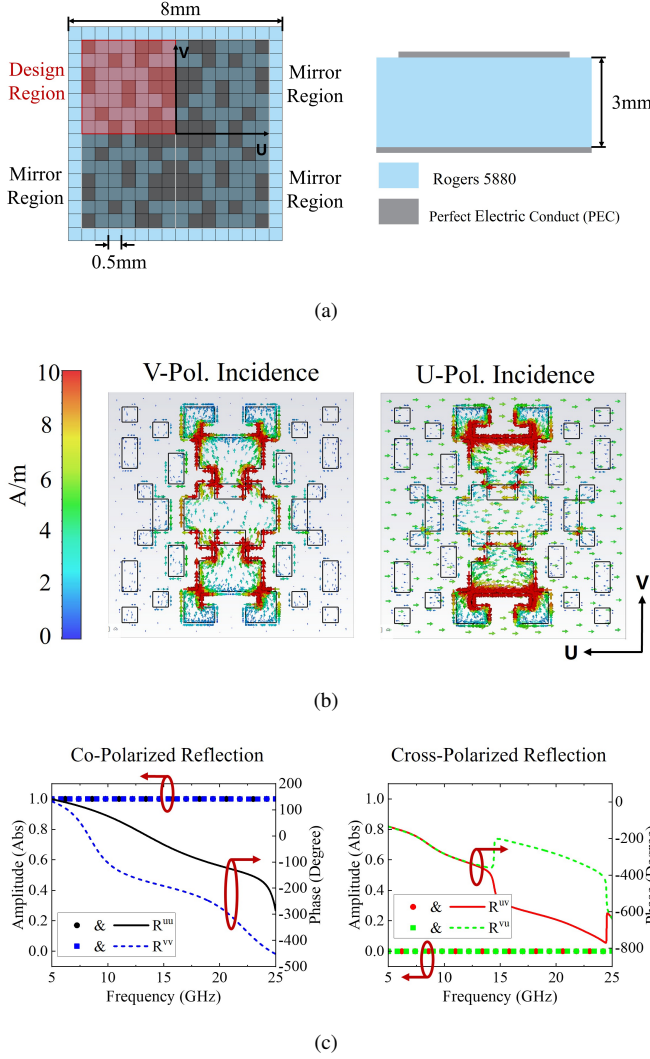


Fig. 3. Schematic diagram of generated metasurface unit. (a) top and side view. (b) current distribution at 15 GHz. (c) Reflection coefficients

exhibit the corresponding mirror-symmetric distributions with respect to the symmetry axes of the unit. The current components contributing to the orthogonally polarized reflection appear in symmetry-related pairs, whose contributions largely cancel each other, thereby suppressing the cross-polarized reflection. Meanwhile, although the overall topology satisfies mirror symmetry, the effective current paths along the U- and V-directions are generally different, resulting in different co-polarized reflection phases $\varphi^{uu/vv}$.

The reflection coefficient of the representative generated mirror-symmetric unit is also presented in Fig. 3(c). It can be observed that the magnitudes of the co-polarized reflection coefficients remain close to 1 over the considered frequency range, whereas the magnitudes of cross-polarized reflection coefficients remain close to 0. In contrast to the relatively stable reflection magnitudes, the co-polarized reflection phases $\varphi^{uu/vv}$ exhibit substantial variations with frequency and are generally different from each other. These results are consistent with the surface-current distributions discussed above and confirm that the generated mirror-symmetric units exhibit non-

polarization-conversion characteristic while retaining different co-polarized reflection phase responses along the U- and V-directions. Based on this characteristic, the surrogate model introduced in the following subsection is designed to predict only the two co-polarized reflection phases $\varphi^{uu/vv}$, thereby reducing the output dimension and learning complexity while retaining the dominant electromagnetic information required for RCS reduction design.

C. The Design of CNN-based Surrogate Model

As shown in Fig. 4, a deep residual network (ResNet [32]) is adopted as the surrogate model to predict the ϕ^{uu} and ϕ^{vv} phase responses of a metasurface unit cell. The architecture consists of a stack of residual blocks, each containing two convolutional layers with batch normalization and ReLU activations, followed by a skip connection that adds the block input to its output. The total number of residual blocks is set to 21, and the hidden dimension is fixed to 64 throughout the network. A linear output layer without activation projects the final hidden representation to the two target phases. The model is trained by minimizing the mean squared error (MSE) between the predicted and the ground-truth phase responses. The surrogate model was implemented and trained in Python 3.11.14, using PyTorch 2.10.0 with CUDA 13.0 support. The training of the surrogate model is performed on a workstation equipped with a 3.90 GHz Intel Xeon Gold 6230 CPU, 512 GB of RAM, and an NVIDIA RTX 3090 GPU.

To train the surrogate model, a total of 4278 unit cell samples are first generated by randomly sampling the design space. Specifically, the initial 4,278 samples are generated by randomly assigning binary values to the 7×7 independent design pixels. Each independent pixel is randomly assigned as '1' (metal) or '0' (air) to construct the pixelated unit-cell topology, while the remaining regions are obtained through the predefined symmetry operations. The corresponding reflection phase ϕ^{uu} and ϕ^{vv} of these generated unit cells are obtained using full-wave simulations in CST Microwave Studio. To further increase the diversity of the dataset, a data augmentation strategy based on electromagnetic symmetry is adopted. A 90° rotation of a metasurface element exchanges its two orthogonal co-polarized reflection-phase responses ϕ^{uu} and ϕ^{vv} . By exploiting this, the dataset is effectively doubled from 4,278 to 8,556 samples, providing a more diverse dataset for surrogate-model training.

The surrogate model was trained for 500 epochs in total. An Adam [33] optimizer was adopted with an initial learning rate of 0.01 and a batch size of 32. A cosine annealing learning rate schedule was employed, decaying the learning rate from an initial value of 0.01 to a final value of 0 over 500 epochs. The dataset contains a total of 8,556 samples, of which 8000 are used for training and 556 for validation. The surrogate model is trained within the frequency range of 5–25 GHz, where stable and accurate phase prediction can be achieved. The phase responses are sampled every 0.03 GHz, resulting in a total of 667 frequency points per phase curve. Since both ϕ^{uu} and ϕ^{vv} phase responses are predicted, the overall output dimension per sample is $2 \times 667 = 1334$.

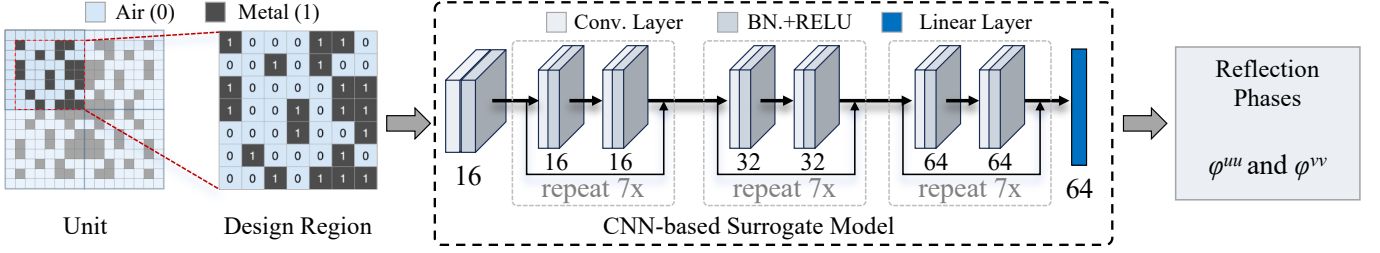


Fig. 4. CNN-based surrogate model takes the binary unit topology as the input and the reflection responses over the target frequency band as the output.

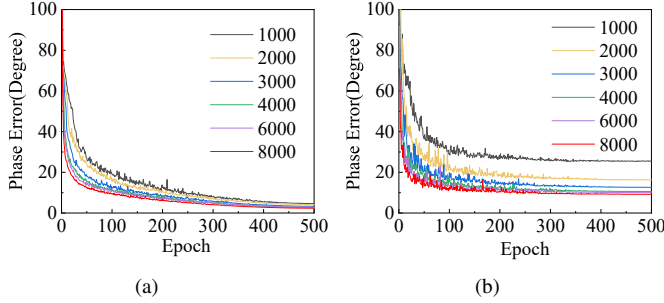


Fig. 5. Phase error of different number of training samples. (a) In training. (b) In validation.

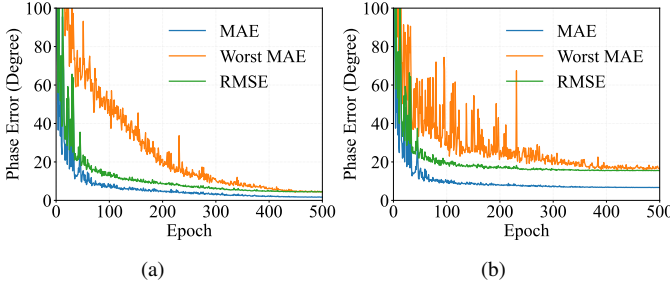


Fig. 6. Phase error of the surrogate model in terms of MAE, RMSE and worst-MAE. (a) In training. (b) In validation.

To investigate the influence of dataset size on the generalization capability of the surrogate model, additional surrogate models were trained using different numbers of training samples, including 1000, 2000, 3000, 4000, 6000, and 8000 samples, while keeping the validation set unchanged. As shown in Fig. 5, the validation phase error gradually decreases with increasing training samples and becomes saturated when the number of training samples approaches 8000. This indicates that the adopted dataset size provides sufficient representative samples for learning the topology-response relationship within the considered pixelated design space. Therefore, 8000 samples are selected for training the final surrogate model, while the remaining 556 samples are used for independent validation.

As shown in Figs. 6 and 7, the phase error of the surrogate model decreased steadily during training. The training MAE converges to approximately 2.38° , indicating stable convergence of the training process. In the validation set, the surrogate model achieves MAE of 9.2° , RMSE of 15.55° ,

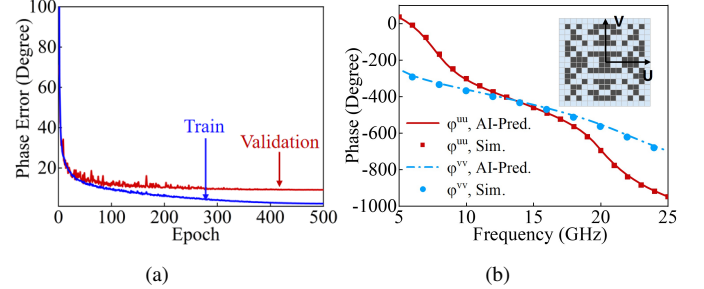


Fig. 7. (a) Phase error of the surrogate model. (b) Comparison between prediction and simulation.

R^2 of 0.989, and a worst-case MAE of 16.75° . To further evaluate the predictive capability, representative examples of the predicted phase responses are compared with full-wave simulation results in Fig. 7(b). The surrogate predictions closely match the simulations across the 5–25 GHz frequency range, demonstrating the reliability of the trained surrogate model.

D. The Design of LSTM-based Policy Model

As shown in Fig. 8, a policy network built on a long short-term memory (LSTM) controller is adopted to generate the full metasurface array configuration. The core of the model is an LSTMCell with a hidden size of 128. At each step, the LSTM receives as input an embedding of the action chosen in the previous step; for the very first step a learnable start token is used instead. The hidden state is then projected onto action logits by one of two task-specific linear heads, both of which take the 128-dimensional hidden state as input: a unit topology head with 2 output units for predicting binary pixel values during the topology generation phase, and an arrangement head with 4 output units for selecting one of the four basic unit types during the array layout phase. The generation proceeds for a total of 214 steps: the first $4 \times 7 \times 7 = 196$ steps produce the four 7×7 binary topologies, and the subsequent $6 \times 6/2 = 18$ steps assign unit indices to one half of the chessboard. The full 6×6 arrangement is then obtained by duplicating each chosen unit with its 90° rotated counterpart, thereby effectively utilizing 8 distinct unit variants in the array.

The policy model is trained for 2000 iterations in total. An Adam [33] optimizer is adopted with a learning rate of $1e-4$ and a batch size of 32. During training, the policy interacts

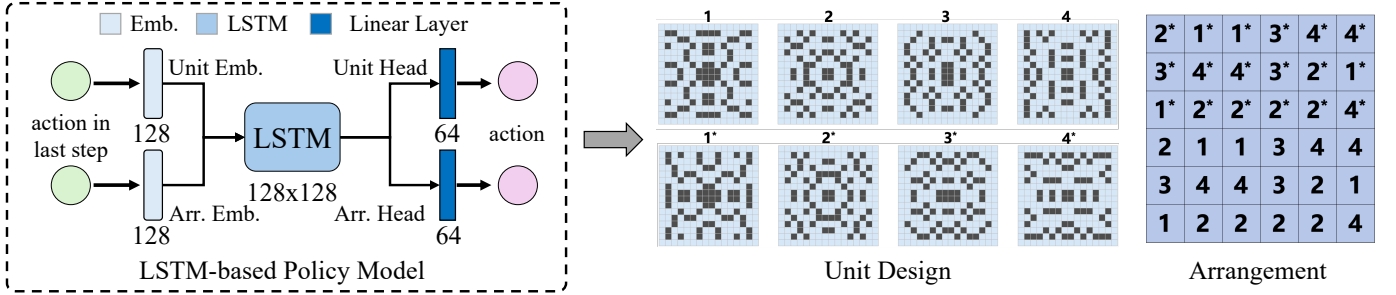


Fig. 8. The LSTM-based policy model sequentially generates the unit design and the corresponding arrangement. At each generation step, the action chosen in the previous step is first embedded and then fed as input to the LSTM for the current step.

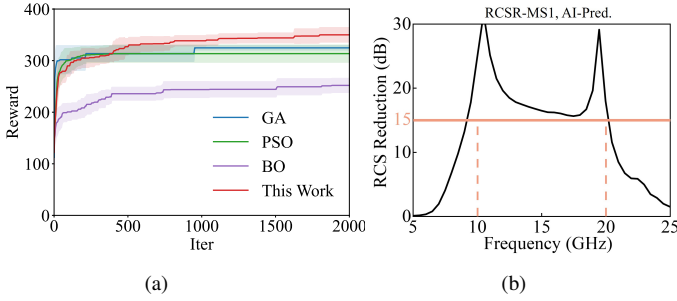


Fig. 9. The results of AI-Driven framework for RCSR-MS1. (a) Reward of different methods per iter. (b) The predicted RCSR results of this work.

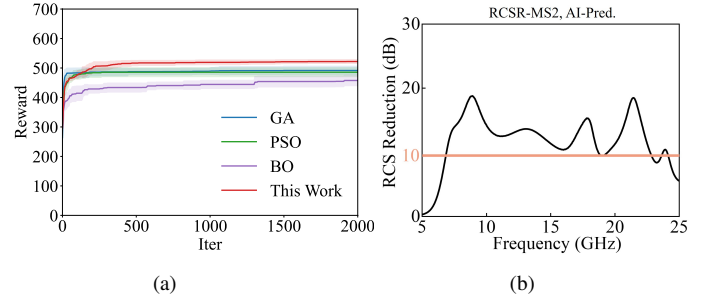


Fig. 10. The results of AI-Driven framework for RCSR-MS2. (a) Reward of different methods per iter. (b) The predicted RCSR results of this work.

with the electromagnetic environment simulated by the surrogate model to collect rollout trajectories, which are then used to update the policy via a policy gradient algorithm with an entropy bonus of 0.001. The policy model was implemented and trained using the same software environment and the same hardware workstation as the surrogate model.

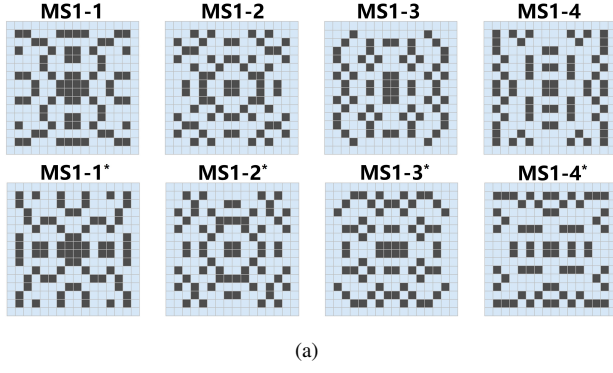
During the training of the LSTM-based policy model, two distinct design cases were considered to demonstrate the flexibility of the proposed framework: RCSR-MS1, which targets a 15 dB RCS reduction over the 10–20 GHz band, and RCSR-MS2, which aims for a broadband 10 dB reduction. These two cases are realized by configuring the reward function parameters accordingly. Once the surrogate model is established, the complete design process for each prescribed RCS reduction case can be completed within approximately 10 minutes.

As shown in Figs. 9(a) and 10(a), the proposed LSTM-based policy optimization is compared with conventional optimization algorithms, including genetic algorithm (GA), particle swarm optimization (PSO), and Bayesian optimization (BO). For a fair comparison, all algorithms are independently evaluated over 10 runs using the same surrogate-assisted evaluation framework and identical evaluation budgets. Therefore, the comparison focuses on the optimization capability of different search strategies under the same computational budget rather than the total cost of electromagnetic simulation. From the results, all compared optimization methods have reached a relatively stable convergence stage under the same number of iterations. Importantly, the training reward of this work outperforms other approaches and increases steadily throughout the optimization for both cases, eventually converging to

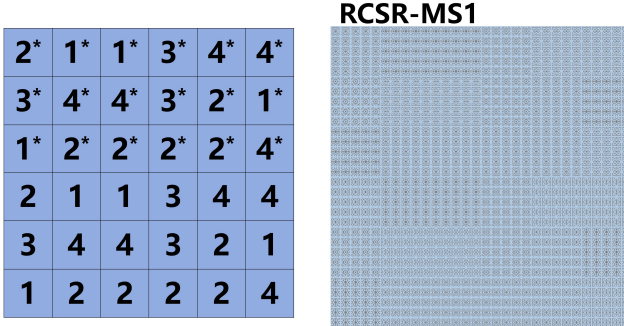
350.4 ± 14.06 and 520.1 ± 6.14 , respectively, indicating stable learning and consistent policy improvement. Although variations exist among different independent training runs due to the stochastic nature of reinforcement learning, the proposed reward optimization strategy in (11) helps alleviate the influence of random fluctuations. After optimization, multiple candidate designs are sampled and the highest reward among the sampled candidates is selected.

It is worth noting that the initial reward values differ between the two cases. This difference arises from the definition of the reward, which counts the maximum number of consecutive frequency points satisfying the RCS reduction target. For RCSR-MS1, with a stringent 15 dB target, the randomly initialized policy model initially satisfies only about 50 points. In contrast, RCSR-MS2 has a more moderate 10 dB target, allowing the initial policy to meet the target at a larger number of frequency points.

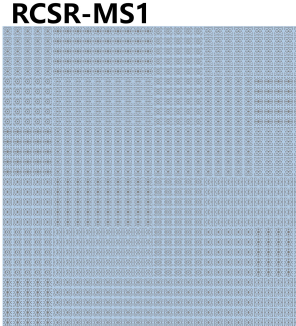
To evaluate the electromagnetic performance of the generated designs, the surrogate-predicted monostatic RCS reduction performance under normal incidence is presented in Figs. 9(b) and 10(b). Owing to the mirror-symmetric unit-cell configuration and the corresponding 90° -paired array arrangement, the generated metasurfaces exhibit polarization-insensitive characteristics under normal incidence. Therefore, the predicted curves represent the normal-incidence RCS reduction performance under the polarization-insensitive design condition, and the corresponding LP and CP RCS reduction characteristics are further verified by the results in Section IV. For RCSR-MS1, the predicted RCS reduction remains above 15 dB over approximately 9.5–20 GHz, satisfying the pre-



(a)



(b)



(c)

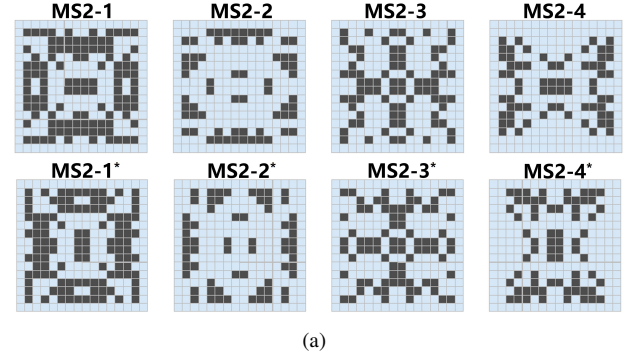
Fig. 11. Schematic of the designed RCSR-MS1. (a) Generated metasurface unit topology, including 4 basic units and their corresponding rotating units. (b) Metasurface units arrangement. (c) Overall layout of the RCSR-MS1.

scribed 15-dB reduction objective over the target band of 10–20 GHz. For RCSR-MS2, the predicted RCS reduction remains above 10 dB over approximately 7–22.5 GHz, satisfying the prescribed broadband 10-dB reduction objective. These results confirm that the proposed framework can automatically generate complete RCS reduction designs according to different prescribed RCS reduction objectives.

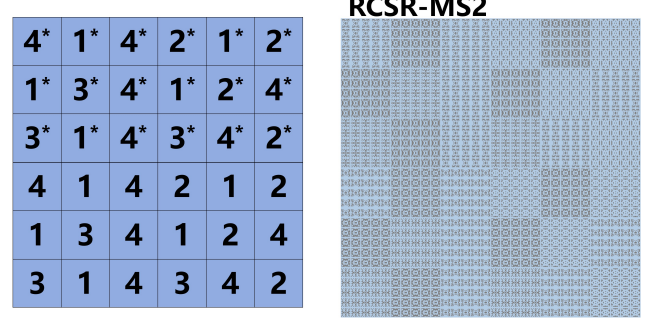
E. The Design of RCS Reduction Plate

Once the policy model is trained, a set of candidate metasurface designs is sampled and evaluated using the surrogate model, and the configuration achieving the highest reward is selected as the final design for each design target. This process yields two optimized metasurfaces, termed RCSR-MS1 and RCSR-MS2, each composed of four distinct pixelated unit cells and their 90° rotated counterparts.

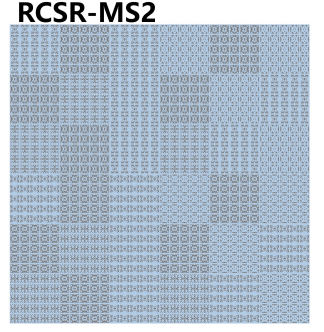
For RCSR-MS1, which targets a 15dB RCS reduction over the 10–20GHz band, the 4 base units (denoted as MS1-1, MS1-2, MS1-3, and MS1-4) are 7×7 binary pixel structures directly generated by the policy network. Based on the paired arrangement Strategy, 90° rotations of these units are obtained, denoted as MS1-1*, MS1-2*, MS1-3*, and MS1-4*, resulting in a total of 8 distinct topologies that are arranged on the 6×6 chessboard-like array according to the predicted assignment. The unit topologies, array layout, and final metasurface configuration are shown in Fig. 11(a)-(c). The surrogate-predicted RCS reduction confirms that the designed metasurface maintains an RCS reduction above 15 dB from approximately 9.5 to 20 GHz, satisfying the design target.



(a)



(b)



(c)

Fig. 12. Schematic of the designed RCSR-MS2. (a) Generated metasurface unit topology, including 4 basic units and their corresponding rotating units. (b) Metasurface units arrangement. (c) Overall layout of the RCSR-MS2.

RCSR-MS2 is designed for a wideband 10 dB RCS reduction. Its 4 basic units (MS2-1, MS2-2, MS2-3, MS2-4) exhibit different pixelated patterns optimized for wideband operation, together with their rotated counterparts (MS2-1*, MS2-2*, MS2-3*, MS2-4*). The corresponding configurations and the predicted RCS reduction are presented in Fig. 12. The results show that an RCS reduction above 10dB is achieved from about 7 to 22.5GHz, covering a broad frequency band.

IV. FABRICATION AND MEASUREMENT

To validate the proposed AI-driven framework, RCSR-MS1 and RCSR-MS2 were simulated, fabricated, and measured. The fabricated prototypes are shown in Fig. 13(a) and (b), and the corresponding measurements were conducted in an anechoic chamber, as illustrated in Fig. 13(c) and (d). Two horn antennas connected to a vector network analyzer (VNA) were employed as the transmitting and receiving antennas, respectively. After system calibration and post-processing of the measured S_{12} data, the RCS results of the fabricated designs were obtained. The specific simulated and measured RCS reduction results are presented and discussed below.

The RCS reduction performances of the generated metasurfaces under normally incident linearly polarized waves are compared in Fig. 14. For RCSR-MS1, the simulated RCS reduction remains above 15 dB over approximately 9.5–21 GHz, while the measured 15-dB RCS reduction bands are 9.1–21.5 GHz and 9.6–21.2 GHz under X- and Y-polarized incidences, respectively. For RCSR-MS2, the simulated results achieve more than 10-dB RCS reduction over approximately

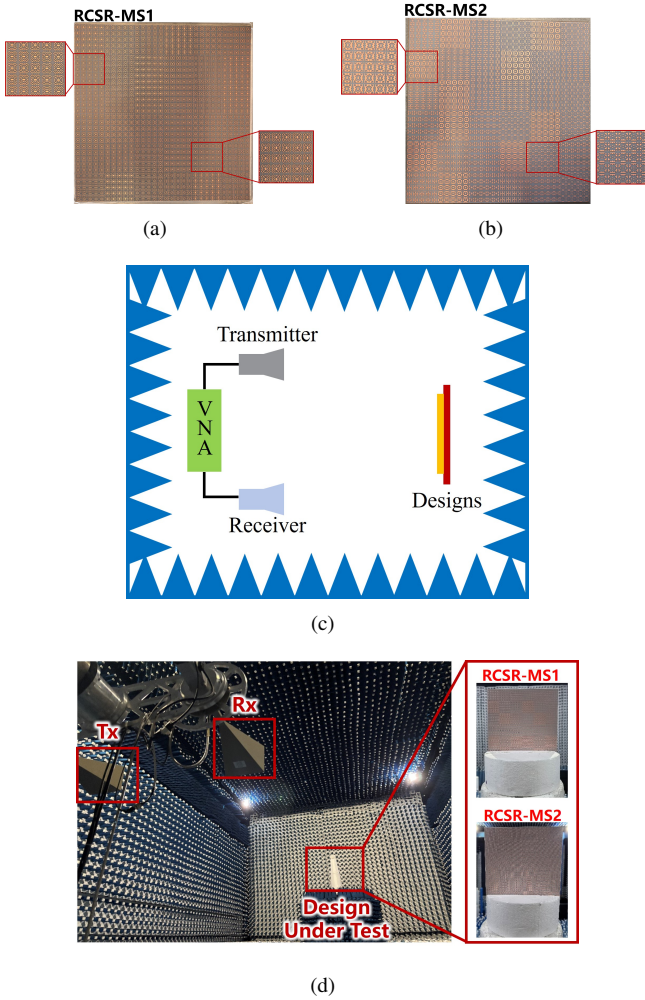


Fig. 13. Fabricated prototype and Measured environment. (a) RCSR-MS1. (b) RCSR-MS2. (c) Schematic of the two-antenna quasi-monostatic measurement configuration. (d) the photo of the actual measurement in the anechoic chamber.

7–23 GHz, whereas the measured 10-dB RCS reduction bands are 7–22.8 GHz (106.1% relative bandwidth) and 7–23.5 GHz (108.2% relative bandwidth) under X- and Y-polarized incidences, respectively. Although some discrepancies exist between the simulated and measured results, likely due to fabrication tolerances and measurement errors, the overall measured responses agree well with the simulations, and both fabricated designs satisfy their respective prescribed RCS reduction objectives. These results validate the effectiveness of the proposed framework in generating complete metasurface designs for different RCS reduction objectives.

Fig. 15 presents the simulated 3-D scattering patterns of the equal-sized PEC reference plate, RCSR-MS1, and RCSR-MS2 at 10, 13.5, 16.5, and 20 GHz under normally incident linearly polarized (LP) and circularly polarized (CP) waves. Compared with the PEC reference plate, the generated metasurfaces exhibit significantly suppressed scattering around the monostatic backscattering direction under X- and Y-polarized as well as LHCP and RHCP incidences, demonstrating their polarization-insensitive scattering characteristics over different frequencies. To enable a direct comparison of the scattering

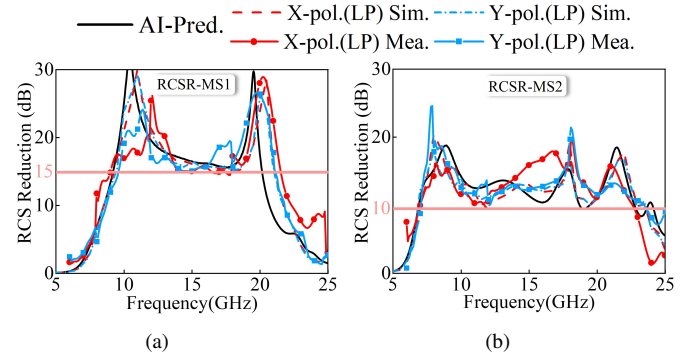


Fig. 14. Comparison of the predicted, simulated, and measured monostatic RCS reduction results under normally incident linearly polarized waves. (a) RCSR-MS1. (b) RCSR-MS2.

levels, the PEC reference plate and the generated designs are presented using the same absolute RCS color scale at each frequency. From the results, although several scattering lobes remain in the proposed designs, comparison under the same absolute RCS color scale shows that their absolute RCS levels are markedly lower than the PEC backscattering peak, while the monostatic backscattering is substantially suppressed. The RCS reduction mechanism of the proposed designs relies on a destructive-interference phase distribution formed by the generated unit cells and their arrangement in the monostatic direction. As a result, the backscattered field in the monostatic direction is significantly suppressed, while part of the scattered energy is redistributed toward other spatial directions. Therefore, significant monostatic RCS reduction can be achieved even though several off-axis scattering lobes remain in the three-dimensional scattering patterns. The consistent scattering-suppression trends observed under different LP and CP incidences further confirm that the generated designs maintain effective monostatic RCS reduction characteristics under different polarization states.

Furthermore, the RCS reduction performance under oblique incidence is evaluated in the corresponding specular reflection direction, and the results at different incidence angles are presented in Fig. 16. As the incidence angle increases, the RCS reduction performance of both RCSR-MS1 and RCSR-MS2 gradually degrades. Nevertheless, for incidence angles up to approximately 30° , the two generated designs generally maintain their prescribed RCS reduction objectives, with only limited degradation over relatively narrow frequency intervals. When the incidence angle further increases to 45° , more noticeable degradation appears over portions of the target frequency bands. The degradation under oblique incidence mainly results from the variation of the unit-cell electromagnetic responses with increasing incidence angle, which weakens the phase-cancellation condition for RCS reduction. Overall, the generated designs exhibit a certain degree of angular robustness under LP incidence.

Then, the simulated RCS reduction responses of RCSR-MS1 and RCSR-MS2 under incident circularly polarized waves are also presented in Fig. 17. Under normal incidence of circularly polarized waves (LHCP and RHCP), RCSR-MS1

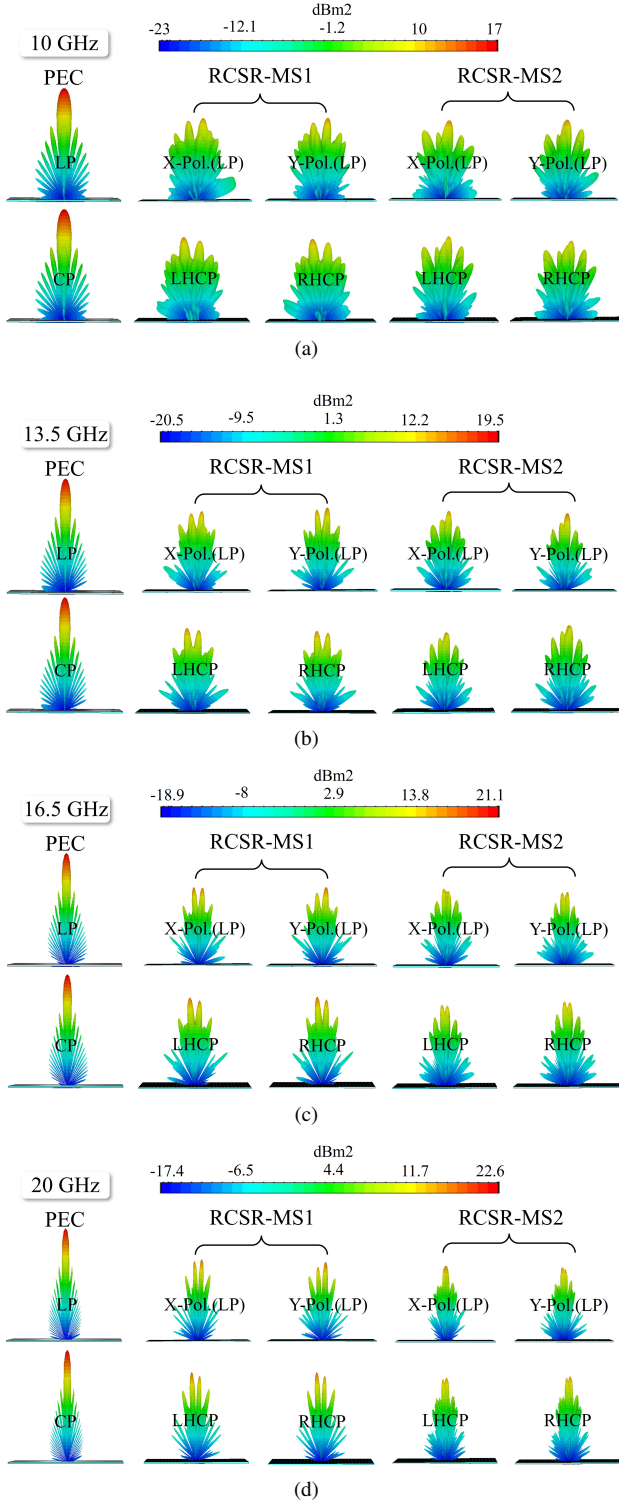


Fig. 15. 3-D scattering patterns of the reference PEC plate and the designed RCSR metasurfaces under normal incidence with different polarizations. (a) 10 GHz. (b) 13.5 GHz. (c) 16.5 GHz. (d) 20 GHz.

maintains more than 15-dB RCS reduction over approximately 9.5–21 GHz, while RCSR-MS2 achieves more than 10-dB RCS reduction over approximately 7–23 GHz. The nearly coincident responses under LHCP and RHCP incidences, together with the closely matched X- and Y-polarized results, demonstrate that the generated metasurfaces can maintain

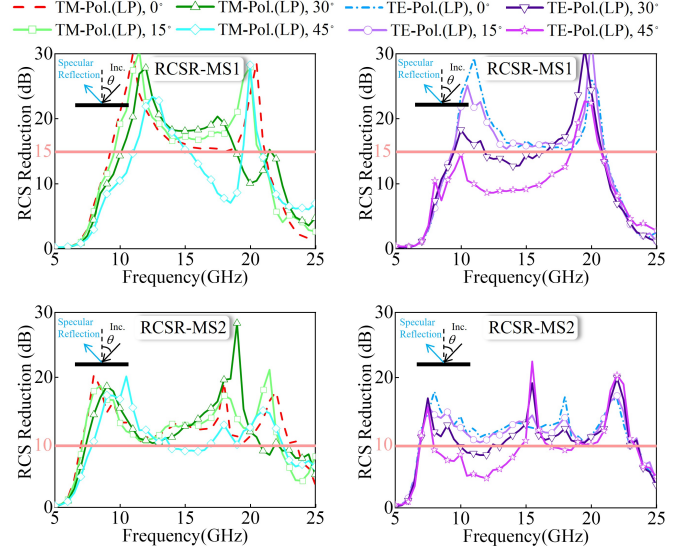


Fig. 16. RCS reduction results of RCSR-MS1 and RCSR-MS2 in the specular reflection direction under linearly polarized waves at different incidence angles.

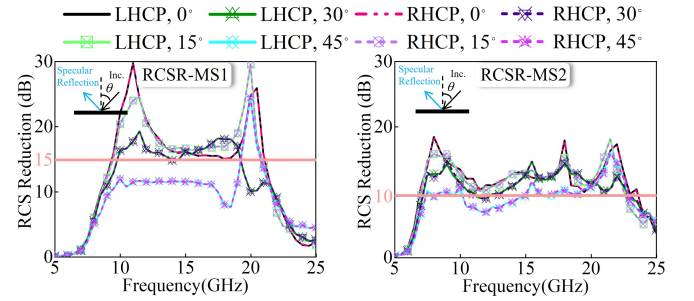


Fig. 17. RCS reduction results of RCSR-MS1 and RCSR-MS2 in the specular reflection direction under circularly polarized waves at different incidence angles.

their prescribed RCS reduction performance under different polarization states at normal incidence. These results further verify the polarization-insensitive characteristics enabled by the mirror-symmetric structure of units and the corresponding paired array arrangement. Moreover, the generated metasurfaces also exhibit a certain degree of angular stability under oblique CP incidence. Although the RCS reduction performance gradually degrades with increasing incidence angle, both RCSR-MS1 and RCSR-MS2 still maintain stable RCS reduction characteristics under LHCP and RHCP incidences for angles up to approximately 30°. More noticeable degradation is observed when the incidence angle increases to 45°. Overall, these results show that the generated designs exhibit polarization-insensitive RCS reduction characteristics together with a certain degree of angular stability, maintaining relatively stable RCS reduction performance under different polarization states for incidence angles up to approximately 30°.

Table I provides a comparison between the proposed designs and other existing designs in terms of design method, structural profile, and electromagnetic performance. As can be observed, most reported approaches still follow a multi-step design

TABLE I
COMPARISON OF REPORTED RCS REDUCTION DESIGNS.

Ref.	Design Method	Substrate Layers / Thickness (mm)	RCSR Bandwidth (GHz, %)		Pol.	Angle Stability	Mechanism
[34]	Expert-Guided Unit Topology Design & Parameter Tuning & Arrangement Design	Dual Sub. / 3.3 mm	(7 dB RCSR) 6.9–17.9 GHz, 88.3%		LP & CP	60°	Phase Cancellation
[26]	Expert-Guided Unit Topology Design & Parameter Tuning & Arrangement Design	Single Sub. / 3 mm	(10 dB RCSR) 7.8–23.2 GHz, 99.3%		LP	60°	Phase Cancellation
[28]	Expert-Guided Unit Topology Design & Parameter Tuning & Arrangement Design	Dual Sub. / 4.75 mm	(10 dB RCSR) 7–28 GHz, 120%		CP	45°	Polarization Conversion + Phase Cancellation
[35]	Expert-Guided Unit Topology Design & Parameter Tuning + Cubic Phase Arrangement Design	Single Sub. / 2.03 mm	(10 dB RCSR) 11–28 GHz, 87.1%		LP & CP	60°	Phase Cancellation
[36]	Expert-Guided Unit Topology Design & Parameter Tuning + GA Arrangement Optimization	Single Sub. / 3 mm	(10 dB RCSR) 9.37–22.79 GHz, 83.5%		LP	45°	Polarization Conversion + Phase Cancellation
[37]	Expert-Guided Unit Topology Design & Parameter Tuning & Arrangement Design	Dual Sub. / 4 mm	(10 dB RCSR) 7.49–32.23 GHz, 124.57%		LP	45°	Absorption + Polarization Conversion + Phase Cancellation
[24]	Expert-Guided Unit Topology Design + Machine-Learning Parameters Optimization	Single Sub. / 1.5 mm	(6 dB RCSR) 13–40 GHz, 101.9%		LP	N.A.	Phase Cancellation
[23]	Expert-Guided Unit Topology Design + Supervised-Learning Parameter Optimization	Single Sub. / 1.5 mm	(10 dB RCSR) 14.8–37.2 GHz, 86%		LP	N.A.	Phase Cancellation
[22]	Expert-Guided Unit Topology Design + TNN Parameter Optimization + DQN Arrangement Optimization	Dual Sub. / 8 mm	(10 dB RCSR) 3.7–18.7 GHz, 133.9%		LP	45°	Absorption + Polarization Conversion + Phase Cancellation
This Work	AI-Driven Automated Design	Single Sub. / 3 mm	RCSR-MS1	(15 dB RCSR) 9.6–21.2 GHz, 75.3%	LP & CP	30°	Phase Cancellation
			RCSR-MS2	(10 dB RCSR) 7–22.8 GHz, 106.1%			

procedure with substantial dependence on expert experience and human intervention. An appropriate unit-cell topology is generally selected based on electromagnetic knowledge, followed by structural-parameter adjustment and subsequent array-arrangement design. Although optimization algorithms or machine-learning techniques can accelerate the latter steps, the initial unit-cell topology generally remains manually pre-defined. In contrast, the proposed framework directly searches for and generates both the unit-cell topologies and their corresponding arrangements according to different prescribed RCS reduction objectives. Therefore, the main distinction of the proposed work lies in its requirement-driven objective-to-complete-design capability, which substantially reduces the dependence on expert-guided topology selection and repeated redesign according to different objectives.

In addition to the design method, the designs generated by the proposed framework still exhibit favorable electro-

magnetic performance. RCSR-MS1 achieves 15-dB RCSR over 9.6–21.2 GHz, while RCSR-MS2 realizes broadband 10-dB RCSR over 7.0–22.8 GHz (106.1% fractional bandwidth) using a single-substrate, 3-mm-low-profile configuration. Compared with other single-substrate and low-profile designs, RCSR-MS2 still demonstrates favorable broadband RCS reduction performance. Besides, the generated designs exhibit polarization-insensitive RCS reduction characteristics under both LP and CP incidences and maintain relatively stable RCS reduction performance for incidence angles up to approximately 30°.

Overall, the proposed framework transforms the conventional expert-dependent and multi-step design procedure into a more direct requirement-to-complete-design process for different prescribed RCS reduction objectives. At the same time, the automatically generated designs still maintain favorable electromagnetic performance, including broadband RCS re-

duction, polarization-insensitive characteristics, angular stability, and low-profile configuration. These results demonstrate the practical advantage of the proposed framework in combining automated complete-design generation with favorable electromagnetic performance.

V. CONCLUSION

An AI-driven framework for diverse monostatic RCS reduction metasurface design has been developed. Different from conventional target-specific design approaches that generally rely on expert-guided unit-cell topology selection, structural-parameter adjustment, and subsequent array-arrangement design, the proposed framework establishes a more direct objective-to-complete-design process that automatically generates both the unit-cell topologies and their corresponding arrangements according to different prescribed RCS reduction objectives. To validate this capability, two RCS reduction designs, denoted as RCSR-MS1 and RCSR-MS2, are generated for different prescribed objectives and subsequently verified through simulations, fabrication, and measurements. The measured results show that RCSR-MS1 achieves 15-dB RCS reduction over 9.6–21.2 GHz, while RCSR-MS2 achieves 10-dB RCS reduction over 7.0–22.8 GHz (106.1% fractional bandwidth). In addition, the generated designs exhibit polarization-insensitive RCS reduction characteristics under both LP and CP incidences and also have a certain degree of angular stability. These results demonstrate the effectiveness of the proposed framework while retaining favorable electromagnetic performance. The proposed framework also shows potential for broader automated metasurface design applications.

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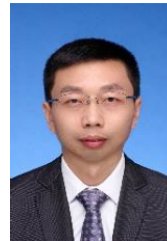
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